

The empirical recurrence for OEIS A297746: recovering a conjecture that is not written down

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Abstract

OEIS A297746 records “Empirical recurrence of order 77”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 129$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 77, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 77 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297746 is “Number of $n \times 5$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1, 3 or 4 neighboring 1s.”. It has offset 1 and begins

6, 71, 452, 3780, 34159, 290114, 2561262, 22815719, ...

The entry states:

Empirical recurrence of order 77 (see link above)

As of the “Last modified” line on the live entry (Jan 05 2018 15:58:45, revision 5) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 129$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 129$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 258$ exact terms of the model returns order 77 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	7	c_{21}	140243468857	c_{41}	−481164836629033	c_{61}	−76757790359421
c_2	5	c_{22}	158257401845	c_{42}	−169485160774687	c_{62}	20025818266934
c_3	368	c_{23}	−590834384728	c_{43}	519144437639697	c_{63}	6057004977880
c_4	−1888	c_{24}	−690332028286	c_{44}	469592885147603	c_{64}	28040548364430
c_5	−671	c_{25}	−804603749543	c_{45}	572110308292142	c_{65}	358513302692
c_6	−50848	c_{26}	6431990534312	c_{46}	−140633287693368	c_{66}	1904693521808
c_7	196254	c_{27}	5503444507203	c_{47}	−374076823403002	c_{67}	−6105394205496
c_8	−64876	c_{28}	−13463606969840	c_{48}	−716455812376401	c_{68}	−4449450561696
c_9	3215628	c_{29}	−19168317358440	c_{49}	−178038125583366	c_{69}	855155164056
c_{10}	−10760454	c_{30}	−27143755074060	c_{50}	13999075793427	c_{70}	−289119800528
c_{11}	9352545	c_{31}	65034448115463	c_{51}	533545841274672	c_{71}	726474996304
c_{12}	−107898196	c_{32}	77239716962728	c_{52}	66107028281728	c_{72}	6994269760
c_{13}	317278597	c_{33}	27251512136968	c_{53}	508646035367522	c_{73}	71574929280
c_{14}	−389116954	c_{34}	−111434702652601	c_{54}	−442077676264383	c_{74}	−13158789760
c_{15}	2044075257	c_{35}	−220100398397173	c_{55}	−61412420300380	c_{75}	−11344193792
c_{16}	−4665748221	c_{36}	60490833469153	c_{56}	−219244370751284	c_{76}	−12826529280
c_{17}	6821638049	c_{37}	116016327573075	c_{57}	1157113499657	c_{77}	3982510080
c_{18}	−22293483242	c_{38}	374294776015141	c_{58}	207447024041935		
c_{19}	22253446653	c_{39}	−51382396679146	c_{59}	−67839884976711		
c_{20}	−27722363586	c_{40}	−342134173039918	c_{60}	46313486083977		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 77, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $77 + 4$ terms removed returns order 77 as well, so $\deg q_{\min} = 77 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $77 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 77 that this sequence satisfies — and that family turns out to have exactly one member.

References

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